FFT-Based Vibration and Power Spectral Density Analysis of IMU Data with In-Flight Motor RPM Synchronization on an FPV Drone

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***Abstract*—Flight controllers on multirotor aircraft rely on a high-rate inertial measurement unit (IMU) to close the attitude control loop. Motor- and propeller-induced vibration injects energy into this signal across a band that shifts with throttle, and an unfiltered controller fights this noise instead of the real motion. We present an end-to-end measurement pipeline that captures 1 kHz IMU data and concurrent motor telemetry on the same microsecond clock, then quantifies the vibration spectrum using FFT, Welch's power spectral density (PSD), and the short-time Fourier transform. An STM32H743 microcontroller acquires accelerometer and gyroscope samples from an MPU9250, requests live per-motor RPM from a Betaflight flight controller over the MSP V2 protocol, and streams both records to an SD card in a self-describing binary format. A MATLAB analyzer aligns the two logs, applies a throttle-derived in-flight mask to the spectral averages, and classifies detected peaks against the motor fundamental $f_0 = \mathrm{RPM}/60$ and its harmonics. The dominant vibration energy was observed to track motor RPM and its harmonics rather than fixed structural frequencies, and the strongest vibration peak occurred near the upper end of the RPM distribution (P90), supporting the use of RPM-tracking dynamic notch filters over static notch configurations. Quantitatively, the median dominant gyroscope frequency (92.3 Hz) corresponds to 5538 RPM, close to the P90 of the in-flight RPM distribution, while the mean RPM (4425) predicts only 73.8 Hz; this offset is consistent with vibration amplitude scaling approximately with RPM^2.**

***Index Terms*—IMU, MPU9250, vibration analysis, FFT, Welch PSD, spectrogram, MSP protocol, Betaflight, dynamic notch filter, FPV drone.**

I. Introduction

Small multirotor aircraft are inherently unstable. The flight controller closes the attitude loop at a rate measured in kilohertz, using the gyroscope as the direct error input of its inner PID stage and the accelerometer to correct long-term drift [1]. The same IMU also senses every mechanical vibration of the airframe. Motors, propellers, and the frame itself excite a band whose dominant frequencies move with throttle, so the noise that contaminates the control signal is non-stationary and tied to the operating point of the vehicle.

Modern flight stacks address this with *dynamic notch* filters whose center frequency tracks the motor RPM. Designing or characterizing such a filter requires the ability to relate measured vibration to the motor state at the exact instant the vibration was sampled. The original course proposal of this project was a single-sensor study: log MPU9250 samples on a drone, apply the FFT and Welch's PSD, and identify motor- and propeller-driven peaks. In the delivered system that scope was extended into an instrumented pipeline that also captures live per-motor RPM from the flight controller, aligns it to the IMU timeline at microsecond accuracy, and performs the spectral analysis with the motor state available as a covariate. With this addition, peaks in the spectrum can be attributed to specific motor speeds rather than only described, enabling attribution of vibration peaks to specific motor operating conditions [2].

The contribution of this work is not a new spectral method but a transparent, end-to-end measurement: we isolate the contribution of each stage of the pipeline—onboard acquisition, lossless binary logging, synchronized motor telemetry, time alignment, and frequency analysis—and report what each adds to the quality of the result. Beyond reporting the measurement, we also address methodological questions that are decisive for any experimental drone log but rarely discussed: how to exclude disarmed-time and idle samples from a Welch average without breaking its assumptions, and how the time-averaged dominant peak relates to the mean motor RPM when vibration energy scales nonlinearly with RPM.

Concretely, the study addresses the following research questions:

- **RQ1.** Can synchronized motor telemetry improve the interpretation of IMU vibration spectra by attributing observed peaks to a known mechanical source?
- **RQ2.** Do the dominant vibration peaks in the gyroscope spectrum correspond to the motor rotational frequency $f_0 = \mathrm{RPM}/60$ and its harmonics, rather than to fixed structural modes?
- **RQ3.** Is a single static notch filter, tuned at a fixed frequency, sufficient for the observed vibration signature, or is RPM-tracking (dynamic) notch filtering required?

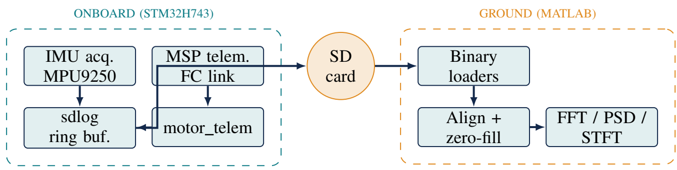

Fig. 1. End-to-end pipeline. The onboard segment acquires IMU and motor telemetry, buffers them in DMA-safe memory, and writes two binary log files. The ground segment loads the files, aligns motor data onto the IMU timeline, and runs the spectral analysis.

The remainder of the paper is organized as follows. Section II describes the hardware platform, the firmware acquisition chain, the motor telemetry protocol, and the spectral analysis. Section III states the experimental conditions. Section IV reports integrity, sensor sanity, frequency-domain, and motor-synchronized findings. Section V interprets the results in the context of dynamic notch design and lists the limitations. Section VI closes with the future work the pipeline now makes possible.

II. Method

A. System Overview

The complete pipeline is shown in Fig. 1. The onboard segment acquires IMU samples and motor telemetry, places them in two independent ring buffers in DMA-safe memory, and flushes them in a bounded, non-blocking task to two files on an SD card. The ground segment (a MATLAB workstation) parses the binary logs, aligns the motor record onto the IMU timeline using the shared microsecond clock, and applies the spectral analysis. The two segments communicate only through the file system: nothing in the measurement chain is processed online.

B. Onboard Hardware

The data logger is built on an STM32H743VIT6 (Cortex-M7, 480 MHz, DevEBox board). The IMU is an MPU9250 read over SPI with the internal digital low-pass filter (DLPF) set to a 92 Hz cutoff, the gyroscope full-scale range to $\pm 2000\,^{\circ}/\mathrm{s}$, and the accelerometer to ± 16 g. The host flight controller is a Zeus F722 running Betaflight 4.5, connected to the logger over UART3 at 230 400 baud. The microSD card is interfaced through SDIO with FATFS [6].

C. IMU Acquisition Pipeline

The MPU9250 sample-ready interrupt schedules a non-blocking SPI DMA read in `Task_1000Hz`. Each completed read is converted to engineering units, timestamped using a software-extended 64-bit counter driven by the DWT hardware cycle counter [5], and pushed as a packed record into the IMU ring buffer. Each record contains a 64-bit microsecond timestamp followed by six single-precision floats (accelerometer and gyroscope, three axes each); the total record size is 32 B. The same monotonic clock is used by the motor telemetry task, so the two streams are aligned without any post-hoc synchronization.

A subtle ordering question arose during integration. If the task first kicks the next SPI read and then logs the previous sample, an unlucky interrupt between the two steps can drop a sample. The actual ordering used is the opposite: the task logs whatever sample is already prepared, and only then kicks the next SPI read. This guarantees that no completed sample is lost between the SPI completion callback and its persistence in the ring buffer.

D. Binary SD Logging (`sdlog`)

A multi-slot binary logger was implemented in C for this work. Each *slot* is an independent data stream with its own ring buffer, record size, and flush policy. The file header is a fixed 32-byte block (`SDLB` magic, version, record size, sample rate, slot name) that makes the file self-describing. The ring buffers reside in non-cacheable AXI SRAM (MPU Region 0) so that DMA writes do not require explicit cache maintenance. A single `Task_50Hz` drains all slots, bounded to 4 KB per tick to avoid blocking the higher-rate tasks. Two slots are used: an IMU slot (1 kHz $\times$ 32 B $\approx$ 32 KB/s) and a motor slot (50 Hz $\times$ 64 B $\approx$ 3.2 KB/s). Byte-perfect on-disk integrity was verified across hundreds of files; an earlier text-mode prototype that exhibited intermittent corruption was abandoned in favor of the binary path.

E. Motor Telemetry over MSP V2

The Multiwii Serial Protocol V2 [3], used by Betaflight as its host interface, was implemented in C from scratch (`msp_lib`). The library performs frame validation with a CRC-8 DVB-S2 checksum, drives a byte-level receive state machine, and enforces a single-outstanding-request model with a configurable timeout watchdog. Each motor record is produced by a two-request cycle in `motor_telem`:

1) `MSP_MOTOR_TELEMETRY`: a 53-byte response carrying, for each of the four motors, the mechanical RPM (`uint32`), the percentage of invalid DSHOT telemetry packets, and the ESC voltage, current, consumption, and temperature.
2) `MSP_MOTOR`: the throttle PWM-equivalent commanded to each motor.

A noteworthy implementation issue concerned the on-wire format of `MSP_MOTOR_TELEMETRY`. The initial parser was written against an outdated specification that described a 7-byte-per-motor record; the current Betaflight source emits 13 bytes per motor. The mismatch was diagnosed by inspecting a recorded `.bin`: motor 0 reported a small positive RPM (DSHOT noise floor at zero throttle), while motors 1–3 reported zeros and impossibly hot temperatures (e.g. 102 °C). Tracing the stride mismatch confirmed that the parser was reading random offsets that landed in the wire payload's zero-padding regions. After the fix, all four motors report consistently.

Each commit produces a 64-byte record containing the timestamp and the per-motor RPM, throttle, ESC voltage, ESC current, ESC temperature, invalid packet percentage, and an MSP flag byte indicating which of the two responses was successfully parsed. The effective record rate is approximately 50 Hz. The bound is not the UART, which carries the round trip in under 4 ms at 230 400 baud, but the flight controller's own MSP servicing latency (approximately 6 ms per round trip in our measurements). Pushing the baud rate higher in pursuit of a faster cycle caused the FC to lock up, so the 230 400 baud configuration was retained.

F. Motor-to-IMU Time Alignment

Because both records share the same microsecond clock, alignment is a one-shot linear interpolation rather than a clock-recovery problem. The MATLAB helper `sdlog_align_motor_to_imu` interpolates every motor field onto each IMU sample time, producing a per-IMU-sample $(\mathrm{RPM}, \mathrm{throttle}, V, I, \ldots)$ tuple at 1 kHz. Zero-valued RPM runs are handled with a length-aware policy: a short run (typically a transient DSHOT dropout of $\leq$ 100 ms) is interpolated across using its flanking nonzero values; longer runs are kept as zero to preserve the genuinely motor-off intervals. A validity mask is returned so that downstream metrics can distinguish real samples from interpolated ones.

G. Spectral Analysis

Three Fourier-domain tools are applied to the gyroscope and accelerometer signals on the IMU-rate timeline. The discrete Fourier transform (DFT) of an N-sample window,

$$X[k] = \sum_{n=0}^{N-1} x[n] \exp\left(-j \, \tfrac{2\pi k n}{N}\right), \tag{1}$$

provides the building block. In practice it is computed with the Cooley–Tukey fast Fourier transform [7], which makes the analysis tractable at the sample counts used here. To obtain a stable, low-variance estimate of the power spectral density we use Welch's method [8], [9]: the signal is divided into overlapping Hann-windowed segments, the magnitude-squared FFT is computed for each segment, normalized by the window energy, and averaged,

$$\hat{P}(f) \; = \; \frac{1}{K} \sum_{k=1}^{K} P_k(f). \tag{2}$$

The Hann window is chosen for its low side-lobe leakage, which is appropriate when narrow tonal components must be separated from a broadband background [10]. To track time variation we use the short-time Fourier transform (STFT) [11], [12] with the same window,

$$S(t, f) \; = \; \left| \sum_{n} x[n] \, w[n-t] \, \exp(-j \, 2\pi f n) \right|^2. \tag{3}$$

The Welch PSD is computed with 1024-sample Hann windows and 50 % overlap, which gives a frequency bin width of approximately 0.98 Hz at the 1 kHz IMU rate. The STFT used for the spectrograms uses a longer 2048-sample Hann window with 50 % overlap, yielding a temporal resolution of about 2.05 s and a frequency resolution of approximately 0.49 Hz. The longer STFT window trades time resolution for a sharper view of narrow spectral lines, which is appropriate when motor RPM changes slowly relative to the window length (motor inertia limits real RPM change to roughly 20 Hz).

H. In-Flight Masking

The drone log starts at power-on but the vehicle is armed and flown only after several seconds of setup. Including the disarmed period in a Welch average dilutes the spectrum: long stretches of near-zero data lower the mean motor-band power and bury narrow peaks. We construct a Boolean mask on the IMU timeline from the synchronized throttle channel and keep only samples for which any motor's commanded throttle exceeds the in-flight threshold (here 1100 μs, just above the idle-armed setpoint). The mask is applied to the PSD inputs and to the vibration-severity health metric but *not* to the spectrogram or the time-domain traces, which retain the full timeline to preserve a useful visual axis and to keep the RPM overlay time-aligned.

I. Harmonic Classification

For each detected gyro peak f_p, candidate motor harmonics are generated from the mean in-flight RPM as

$$f_0 \; = \; \mathrm{RPM}/60, \qquad \mathrm{BPF}_B \; = \; B \cdot f_0,$$

where B is the blade count. The analyzer searches harmonics $\{m f_0\}_{m=1}^{6}$ and $\{m \cdot \mathrm{BPF}_B\}_{m=1}^{6}$ and labels a peak as motor-driven if it lies within $\pm 8\,\%$ of any candidate. The tolerance was chosen empirically to accommodate the spread of motor RPM within each averaging window—the in-flight RPM distribution of the test run varies by approximately $\pm 30\,\%$ around its mean (P10 3185, P90 5885 against a mean of 4425)—while still keeping unrelated mechanical modes from being absorbed into the motor-driven class. A narrower tolerance would miss peaks driven by the upper or lower tails of the RPM distribution; a wider one would over-attribute energy to motor harmonics and obscure structural modes.

III. Experimental Setup

All measurements reported here were obtained on the integrated platform. A representative test run is used throughout the Results section: a 78 s bench-style run with all four motors active, spinning at a mean of 4425 RPM (P10–P90: 3185–5885). The IMU log (`IMU_1000Hz_LOG_003.bin`) contains 74 295 records, and the companion motor log (`MOTOR_100Hz_LOG_003.bin`) covers 77.68 s. The two logs overlap for 77.68 s. The flight-only mask retains 50 313 of 78 000 uniform-grid samples (64.5 %). All spectral analysis uses Welch's method with a Hann window and 50 % overlap; the spectrogram uses the same window over short segments stepped across the full log.

IV. Results

A. Log Integrity

Across 78 s of acquisition the effective IMU rate is 952.5 Hz against the nominal 1000 Hz; 5.25 % of samples are missed. The longest single gap is 25.4 ms, the timestamp sequence is strictly monotonic, and no duplicates appear. Gaps cluster at the SD-card flush windows rather than at random points in the log, which is consistent with write-side back-pressure rather than acquisition jitter; this matches independent measurements with two SanDisk Ultra cards. The motor log carries no timeouts, no CRC errors, and no unexpected responses across the same window.

B. Sensor Sanity

With the vehicle at rest in the first five seconds of the log, the accelerometer reports a gravity vector of $[-0.024,\ +0.008,\ +0.993]$ g (norm 0.994 g, expected 1.000 g) and a per-axis standard deviation of $[0.014,\ 0.025,\ 0.050]$ g. The gyroscope residual bias is $[-0.014,\ 0.000,\ -0.014]$ rad/s and its noise floor standard deviation is $[0.002,\ 0.004,\ 0.002]$ rad/s, consistent with the MPU9250 datasheet [4]. Neither sensor saturates at any point in the log.

C. Vibration Severity and Flight-Only Filtering

The flight-only mask shifts the vibration-severity figures upward in every metric: the gyroscope RMS rises from 0.125 to 0.147 rad/s ($+17\,\%$), the accelerometer RMS from 0.716 to 0.817 g ($+14\,\%$), and the motor-band gyro power roughly doubles ($+54\,\%$). The interpretation is straightforward: the unmasked average had been pulled down by the long disarmed prefix of the log, and the in-flight portion is in fact closer to the (saturated) high end of the severity scale.

D. Gyroscope and Accelerometer Spectra

Figure 2 shows the gyroscope spectrogram with the synchronized motor RPM overlay. Two visual cues drive the interpretation. First, hot pixels in the spectrogram track the overlaid lines f_0, $2f_0$, and $3f_0$ rather than sitting at fixed frequencies; the dominant gyroscope vibration is therefore motor-driven and shifts with throttle. Second, the energy is concentrated below approximately 250 Hz, with a broad lobe at 70–90 Hz and a second cluster at 180–230 Hz. The lower lobe falls inside the in-flight RPM fundamental band ($f_0 \in [53, 98]$ Hz between P10 and P90), and the upper cluster is consistent with the second and blade-pass harmonics of typical hover RPM.

The accelerometer spectrogram (Fig. 3) shows similar harmonic tracking. The role of the two sensors, however, is different: the gyroscope feeds the PID inner loop and its spectrum is what filter design targets, while the accelerometer reflects linear motion of the airframe and so is the diagnostic for mechanical issues—frame resonance, propeller imbalance, and motor-mount stiffness—rather than for filter design itself.

E. Notch Candidate Analysis

Figure 4 plots the summed gyroscope PSD on a logarithmic scale up to the Nyquist limit, with two pairs of vertical guides: dashed lines at the P5 and P95 of the in-flight motor fundamental $f_0 = \mathrm{RPM}/60$ (labelled *RPM* f_0 *p5* and *RPM* f_0 *p95* in the figure), and dotted lines at the frequencies that contain $95\,\%$ and $99\,\%$ of the cumulative gyroscope power. Two interpretive observations follow. Broad lobes that lie inside the RPM band require a dynamic notch, because their center frequency moves with throttle; narrow stationary peaks outside the RPM band would be fixed-notch candidates. The $95\,\%$ cumulative-power line at 207.5 Hz also provides a defensible cutoff for the gyroscope low-pass filter, as content above it contributes a negligible fraction of the total power.

The dominant spectral energy of the gyroscope signal remains inside the RPM-dependent region of the spectrum, rather than concentrating at fixed frequencies that would correspond to structural modes of the airframe. The implication is direct: a single static notch tuned at any one frequency would attenuate only the fraction of the energy that happens to coincide with its center frequency at a given moment, and would leak energy through whenever the motor RPM moves the harmonic away from that center. An RPM-tracking (dynamic) notch—in essence a parameter-adapted filter whose center frequency is steered by an external observation, in the spirit of adaptive filtering [13]—is therefore more appropriate than a static notch configuration for this airframe.

F. Time-Averaged Dominant Peak versus Mean RPM

A specific quantitative finding emerges from comparing the time-averaged dominant peak against the motor state. The mean in-flight RPM of 4425 predicts a fundamental at

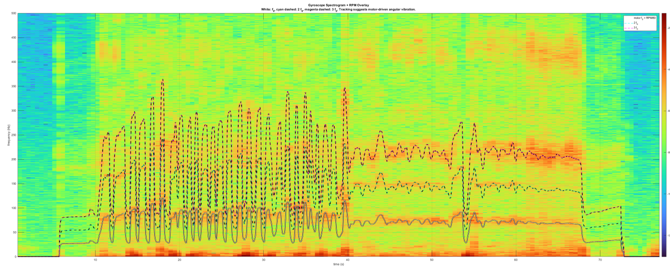

Fig. 2. Gyroscope spectrogram showing the evolution of vibration energy over time. The overlaid solid line is the per-sample motor fundamental $f_0 = \mathrm{RPM}/60$; the dashed lines are $2f_0$ (cyan, two-blade) and $3f_0$ (magenta, three-blade). Dominant vibration bands track motor speed changes rather than remaining at fixed frequencies, demonstrating that the energy is motor-driven and that an RPM-tracking notch is appropriate.

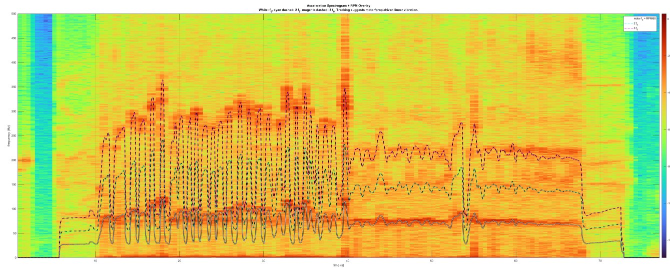

Fig. 3. Accelerometer spectrogram showing the evolution of linear-axis vibration energy over time. The same RPM overlay is shown for reference. Whereas the gyroscope spectrum is the relevant input for controller filter design (since the gyro feeds the PID inner loop), the accelerometer view is a mechanical diagnostic that exposes frame resonance, propeller imbalance, and motor-mount behavior.

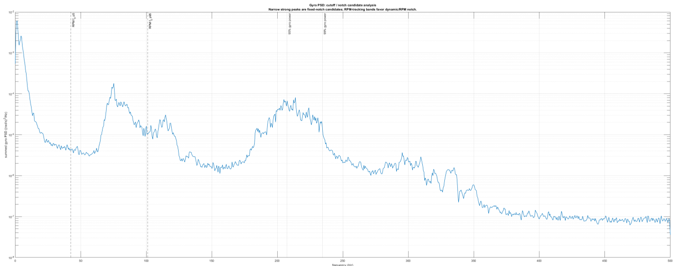

Fig. 4. Summed gyroscope PSD with notch-design guides. Dashed verticals mark the in-flight motor fundamental at the P5 and P95 of the RPM distribution (labelled *RPM* f_0 *p5* and *RPM* f_0 *p95*); dotted verticals mark 95 % and 99 % cumulative gyroscope power. Broad energy inside the RPM band drives a dynamic notch; the 95 % line is a defensible low-pass cutoff.

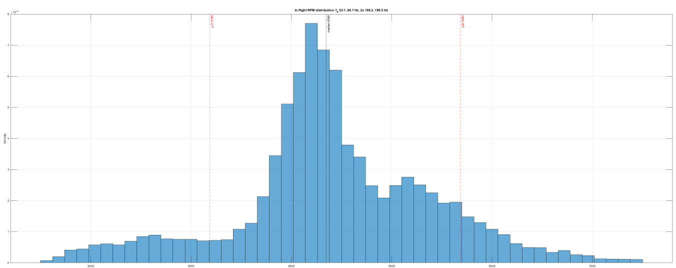

Fig. 5. In-flight RPM histogram. Vertical dashed lines mark the P10 (3185 RPM) and P90 (5885 RPM) of the distribution. The median dominant gyroscope frequency, converted back to RPM ($f_{\text{dom}} \cdot 60 = 5538$ RPM), falls near the P90 rather than at the mean, illustrating the bias of time-averaged spectra toward high-RPM moments.

$f_0 = 73.8$ Hz, yet the median dominant gyroscope frequency across the spectrogram windows is 92.3 Hz. Converting back, $92.3\,\text{Hz} \cdot 60 = 5538$ RPM, which lies very close to the P90 of the RPM distribution (5885). Figure 5 illustrates this graphically: the in-flight RPM distribution is shown with dashed verticals at its P10 (3185 RPM) and P90 (5885 RPM). The RPM value implied by the observed dominant gyroscope frequency (5538 RPM, 92.3 Hz) falls near the P90 of the distribution rather than at its mean (4425 RPM, 73.8 Hz).

The observed behaviour is consistent with vibration amplitude scaling approximately with RPM^2 (lift and unbalance forcing both rise with the square of rotation rate). Under such scaling, the spectrogram windows that sit at high RPM dominate the "loudest peak per window" statistic and pull its median away from the mean. The implication for filter design is direct: a static notch tuned to the arithmetic mean RPM would miss the loudest peaks of the log, and an RPM-tracking notch is required. This result demonstrates the value of synchronized motor telemetry for interpreting vibration spectra.

G. Synthesis

Table I collects the figures of merit for the test run. The firmware-side numbers reflect the actual measurement chain, and the analysis-side numbers reflect what the chain reveals once the data is in hand.

TABLE I
FIGURES OF MERIT FOR THE REPRESENTATIVE TEST RUN.

Quantity	**Value**
IMU nominal / effective rate	1000 Hz / 952.5 Hz
Missed IMU samples	5.25 %
Motor record rate	≈ 50 Hz
Motor MSP timeouts / CRC errors	0 / 0
Test duration (overlap)	77.68 s
Mean RPM (in-flight, all motors)	4425 (P10–P90: 3185–5885)
Predicted f_0 from mean RPM	73.8 Hz
Median dominant gyro freq.	92.3 Hz $\approx$ P90 f_0
Flight-only sample fraction	50 313/78 000 (64.5 %)
Gyro motor-band power (full / flight)	2.5×10^{-3} / 3.9×10^{-3} $(\text{rad/s})^2$
95 % cumulative gyro power freq.	207.5 Hz

V. Discussion

The results expose a coherent picture and a small number of clear limitations.

Advantages.: The measurement chain is fully reversible to the bit on disk and time-aligned to the microsecond. Each stage—acquisition, buffering, transport, persistence, ground-side decoding, and spectral analysis—can be inspected in isolation. The addition of motor telemetry turns descriptive spectra into causal ones: peaks now have an attribution, not just a frequency. The flight-only mask was decisive in practice; without it the motor-band gyroscope power was lower by a factor of approximately 1.5 (understated by about a third), and the saturated vibration-severity score in the masked case is a better summary of the in-flight regime than the unmasked average that mixed flight with idle.

Limitations.: The sampling rate caps the resolvable band at the Nyquist limit of 500 Hz. For the present payload, motor- and prop-driven energy lives below approximately 250 Hz, so 1 kHz is sufficient; however, higher-frequency structural modes are not visible. The motor record rate of 50 Hz is well above the Nyquist limit for the RPM signal itself—motor inertia limits real RPM change to roughly 20 Hz—but it is not enough to follow the gyroscope's 1 kHz spectrogram window-by-window without interpolation. The peak classifier uses the mean in-flight RPM as its fundamental reference; as Section IV-F shows, the time-averaged dominant peak is biased toward the upper end of the RPM distribution, so a classifier that uses the per-window RPM rather than the mean is the natural next step.

Generality.: The result that time-averaged spectra are biased toward high-RPM moments—consistent with vibration amplitude scaling approximately with RPM^2—is a property of the physics of rotating machinery rather than of this particular drone. The same effect should appear on any vehicle whose dominant vibration sources scale with operating point, which makes the in-flight masking and per-window RPM strategies developed here applicable beyond the FPV setting.

VI. Conclusion

We presented a complete IMU vibration measurement pipeline for a multirotor aircraft and quantified it end to end. The original proposal of recording IMU data and applying the FFT was met and extended into a synchronized two-sensor system: 1 kHz IMU acquisition on an STM32H743 alongside live per-motor RPM telemetry from a Betaflight flight controller, both written to SD card as self-describing binary records sharing a common microsecond clock, and a MATLAB analyzer that aligns the two logs and applies Welch's PSD and the spectrogram with motor RPM as a covariate.

The findings follow a single thread: *the spectrum is interpretable only in the context of the motor state*. The dominant gyroscope energy lives in a band that moves with throttle and therefore demands an RPM-tracking dynamic notch rather than a static one; the time-averaged loudest peak is biased toward the upper end of the RPM distribution, an observation consistent with vibration amplitude scaling approximately with RPM^2; and disarmed intervals—unavoidable in any real log that begins at power-on—must be excluded from the spectral averaging if the result is to reflect flight rather than the bench.

The major contributions of this work are:

- a dual-stream binary data logger on the STM32H743 with microsecond-accurate cross-stream synchronization;
- a from-scratch MSP V2 client (`msp_lib`) that acquires live per-motor RPM, throttle, and ESC telemetry from a Betaflight flight controller;
- a synchronized IMU–motor logging architecture in which both streams share the same monotonic clock and are stored in a self-describing binary format;
- a MATLAB motor-aware spectral analysis toolchain combining alignment, smart zero-fill, throttle-derived in-flight masking, Welch PSD, the short-time Fourier transform, and harmonic classification; and
- experimental evidence of RPM-dependent vibration behaviour on the test airframe, including the observation that the time-averaged dominant peak is biased toward the upper end of the RPM distribution.

These contributions directly address the research questions of Section I. **RQ1**: motor telemetry made the spectrum interpretable by attributing peaks to a known mechanical source. **RQ2**: the dominant gyroscope peaks tracked the motor fundamental and its harmonics rather than fixed structural modes. **RQ3**: a static notch is insufficient for this airframe because the dominant energy moves with throttle—an RPM-tracking dynamic notch is required.

Future work makes direct use of the synchronized log: per-motor RPM overlays in place of the four-motor mean, time-resolved blade-pass tracking, RPM-gyro coherence analysis, and an offline simulation of an RPM-tracking dynamic notch that quantifies the achievable attenuation on the same recorded data. An upgrade of the acquisition chain to FIFO-based 8 kHz sampling would also extend the resolvable band to where higher-order structural modes can be observed.

References

[1] S. Bouabdallah and R. Siegwart, "Full Control of a Quadrotor," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, San Diego, CA, USA, 2007, pp. 153–158.

[2] Betaflight Development Team, "RPM Filter Documentation," Betaflight Wiki, 2024. [Online]. Available: https://betaflight.com/docs/

[3] Multiwii Project and Betaflight, "MultiWii Serial Protocol (MSP) V2 Specification," 2024. [Online]. Available: https://github.com/iNavFlight/inav/wiki/MSP-V2

[4] InvenSense, "MPU-9250 Product Specification," Revision 1.1, Document PS-MPU-9250A-01, 2014.

[5] STMicroelectronics, "RM0433: STM32H742, STM32H743/753 and STM32H750 Value Line Advanced Arm-Based 32-Bit MCUs Reference Manual," Revision 8, 2022.

[6] C. Hagihara (ChaN), "FatFs — Generic FAT Filesystem Module," 2023. [Online]. Available: http://elm-chan.org/fsw/ff/

[7] J. W. Cooley and J. W. Tukey, "An Algorithm for the Machine Calculation of Complex Fourier Series," *Math. Comput.*, vol. 19, no. 90, pp. 297–301, 1965.

[8] P. D. Welch, "The Use of the Fast Fourier Transform for the Estimation of Power Spectra: A Method Based on Time Averaging Over Short, Modified Periodograms," *IEEE Trans. Audio Electroacoust.*, vol. 15, no. 2, pp. 70–73, 1967.

[9] P. Stoica and R. Moses, *Spectral Analysis of Signals*. Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2005.

[10] F. J. Harris, "On the Use of Windows for Harmonic Analysis with the Discrete Fourier Transform," *Proc. IEEE*, vol. 66, no. 1, pp. 51–83, 1978.

[11] J. B. Allen and L. R. Rabiner, "A Unified Approach to Short-Time Fourier Analysis and Synthesis," *Proc. IEEE*, vol. 65, no. 11, pp. 1558–1564, 1977.

[12] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1999.

[13] B. Widrow and S. D. Stearns, *Adaptive Signal Processing*. Englewood Cliffs, NJ, USA: Prentice Hall, 1985.